mshqc: A C++20 Quantum Chemistry Engine for Orbital-Optimized Møller–Plesset Perturbation Theory

Muhamad Syahrul Hidayat (Department of Physics, Universitas Gadjah Mada)

15 October 2026

Summary

`mshqc` is a high-performance quantum chemistry library that implements a focused set of wave-function-based electronic structure methods in a compact C++20 core with a low-overhead Python interface built on `nanobind` (Jakob 2022). The library provides restricted, unrestricted, and restricted open-shell Hartree–Fock references (`RHF`, `UHF`, `ROHF`) (Roothaan 1951, 1960; Pople and Nesbet 1954) accelerated by Pulay's Direct Inversion in the Iterative Subspace (`DIIS`) (Pulay 1980, 1982) and Superposition of Atomic Densities (`SAD`) initial guesses, alongside second- and third-order Møller–Plesset perturbation theory in both canonical (`RMP2`, `UMP2`, `RMP3`, `UMP3`) (Møller and Plesset 1934; Pople, Binkley, and Seeger 1976; Head-Gordon, Pople, and Frisch 1988) and orbital-optimized forms (`OMP2`, `OMP3`) (Neese et al. 2009; Bozkaya et al. 2011; Bozkaya 2011).

To eliminate steep memory and computational bottlenecks in four-center electron repulsion integrals (ERIs) evaluated via `libcint` (Sun 2015), `mshqc` unifies three integral representations across all implemented SCF and correlated solvers: exact four-center tensors (`exact`) with Schwarz prescreening (Häser and Ahlrichs 1989), pivot-based Cholesky decomposition (`cholesky`) (Beebe and Linderberg 1977; Koch, Sánchez de Merás, and Pedersen 2003), and auxiliary-basis Density Fitting / Resolution of the Identity (`df`) (Feyereisen, Fitzgerald, and Komornicki 1993; Weigend 2002). High-dimensional tensor contractions in the correlated kernels are dispatched to `TBLIS` (Matthews 2017), while orbital rotations in `OMP2` and `OMP3` are driven by a matrix-free Steihaug–Toint trust-region Second-Order SCF (`SOSCF`) optimizer.

Statement of need

Orbital-optimized Møller–Plesset perturbation methods (`OMP2` and `OMP3`) relax the reference orbitals in the presence of dynamical electron correlation, restoring Hellmann–Feynman satisfaction for one-electron properties, eliminating artificial spin contamination in open-shell radicals, and mitigating first-order orbital relaxation errors (Neese et al. 2009; Bozkaya et al. 2011; Bozkaya 2011). However, because each macro-iteration of an orbital-optimized calculation requires re-transforming two-electron integrals and re-evaluating $\mathcal{O}(N^5)$ (`OMP2`) or $\mathcal{O}(N^6)$ (`OMP3`) amplitude and response contractions, naive implementations quickly become memory- and compute-bound.

Researchers and students who wish to prototype new orbital-optimization algorithms, three-index tensor factorizations, or custom contraction schedules face a steep barrier to entry in large production suites. General-purpose ecosystems are designed for broad method coverage, which necessarily introduces deep abstraction layers between the high-level Python driver and the low-level numerical kernels. `mshqc` addresses this need by providing a compact, inspectable C++20 codebase where the SCF engines, three-index `cholesky`/`df` integral transformers, `TBLIS` tensor contractions, and trust-region `SOSCF` orbital optimizers are unified under a clean, two-tier API. Users can execute complete multi-method pipelines in a single Python call via `mshqc.run()`, or instantiate individual C++ classes directly to inspect and modify intermediate tensors.

State of the field

Mature open-source quantum chemistry packages such as `Psi4` (Smith et al. 2020) and `PySCF` (Sun et al. 2020) offer comprehensive electronic structure capabilities, including canonical and orbital-optimized perturbation theory. In `Psi4`, orbital-optimized methods are implemented within the compiled `fnocc` and `occ` modules, whereas `PySCF`

combines C-level integral evaluation with Python/NumPy tensor operations. While both ecosystems excel as general-purpose production tools, modifying their internal integral transformation pipelines or orbital Hessian preconditioners requires navigating hundreds of thousands of lines of code.

Conversely, educational or lightweight quantum chemistry projects typically implement only canonical `RHF` and `RMP2` in pure Python, lacking open-shell references (`UHF`/`ROHF`), third-order correlation (`MP3`/`OMP3`), point-group symmetry reduction, or rank-reduced three-index integrals (`cholesky` and `df`). `mshqc` bridges the gap between minimalist teaching codes and monolithic production suites: it maintains a small, single-purpose C++20 codebase while delivering execution speeds that match or exceed production packages on `OMP2` and `OMP3` workloads.

Software design

`mshqc` is organized into modular C++20 layers exposed to Python through `nanobind` (Jakob 2022) with zero-copy `Eigen` (Guennebaud, Jacob, et al. 2010) matrix interoperability and automatic Global Interpreter Lock (`GIL`) release during compute-intensive calls:

1. **Molecular Symmetry & Basis Infrastructure (`Molecule`, `BasisSet`, `PointGroup`, `PetiteList`):** Parses Gaussian basis set (`.gbs`) files bundled directly within the Python package—including the Dunning correlation-consistent families (`cc-pVXZ`) (Dunning 1989) and matched `RI`/`JKFIT` auxiliary basis sets (Weigend 2002)—with automatic contraction normalization (Helgaker, Jørgensen, and Olsen 2000). Automated point-group detection aligns molecular coordinates to canonical symmetry axes and constructs the `PetiteList` of symmetry-unique shell pairs (Atkins, Child, and Phillips 1970).
2. **Unified Integral Engine (`IntegralEngine`, `Screening`, `CholeskyERI`, `DensityFittingERI`):** Evaluates analytical one- and two-electron integrals in spherical harmonic or Cartesian representations via `libcint` (Sun 2015). All solvers accept a unified `eri_method` configuration (`"exact"`, `"cholesky"`, or `"df"`), reducing two-electron storage scaling from $\mathcal{O}(N^4)$ four-center tensors to $\mathcal{O}(N_{\text{aux}}N^2)$ three-center metric-decomposed matrices B_{pq}^{Q} (with optional `HDF5`-backed out-of-core disk storage (The HDF Group 1997)).
3. **Self-Consistent Field (`RHF`, `UHF`, `ROHF`):** Shares a common base infrastructure (`BaseSCF`) supporting SAD initial guesses, Schwarz screening, symmetry-reduced Fock construction, and `DIIS` extrapolation. For open-shell systems under Density Fitting, exchange contractions exploit occupied-orbital rank projection to avoid $\mathcal{O}(N_{\text{aux}}N^3)$ bottlenecks.
4. **Correlated & Orbital-Optimized Solvers (`RMP2`, `UMP2`, `OMP2`, `RMP3`, `UMP3`, `OMP3`):** Dispatches four-index tensor contractions to `TBLIS` (Matthews 2017) and three-index `df`/`cholesky` contractions to cache-blocked `OpenMP` + `BLAS` (`DGEMM`/`DSYRK`) kernels (Dagum and Menon 1998). In `OMP2` and `OMP3`, non-redundant orbital rotation parameters κ_{ai} are optimized using a matrix-free Steihaug–Toint truncated conjugate gradient (`TrustRegionSOSCF`) solver coupled with unrelaxed and Z-vector response densities.

Validation and performance

All methods in `mshqc` (`RHF`/`UHF`, `RMP2`/`UMP2`, `OMP2`, and `OMP3`) have been benchmarked against `Psi4` (Smith et al. 2020) across closed-shell (`Ne`, H_2O) and open-shell (`Li`, `C`, `O`) systems using the `cc-pVTZ` basis set (`cc-pVTZ-RI` auxiliary basis for `df`, all electrons correlated, decomposition and convergence thresholds set to 10^{-9} Ha). Across all 60 benchmark calculations (`5 systems × 3 ERI modes × 4 methods`), `mshqc` reproduces `Psi4` reference energies to within $|\Delta E| \leq 1.27$ μHa (with 54 of 60 runs matching to 0.00 μHa at 10^{-8} Ha precision, and all 20 `df` runs matching `Psi4` to $|\Delta E| \leq 0.01$ μHa).

As shown in Figure 1 and Table 1, `mshqc` outperforms `Psi4` in all 60 test cases, achieving wall-clock speedups of 2.4× to 18.6× on `OMP2` and 2.8× to 11.6× on `OMP3`, while the `TrustRegionSOSCF` optimizer converges the orbital gradient norm $\|\mathbf{g}\|_2$ below 10^{-5} Ha within 3 to 4 macro-iterations across all tested systems. On H_2O ($N = 58$, $N_{\text{aux}} = 141$), switching from four-center `exact` integrals to three-center `cholesky` and `df` integrals in `mshqc` monotonically reduces the `OMP3` wall-clock time from 0.797 s (`exact`) to 0.705 s (`cholesky`) and 0.358 s (`df`).

Table 1: Representative benchmark comparison between `mshqc` and `Psi4` (`cc-pVTZ` / `cc-pVTZ-RI`, threshold 10^{-9} Ha), reporting converged total energy, absolute deviation $|\Delta E| = |E_{\text{mshqc}} - E_{\text{Psi4}}|$ in μHa, wall-clock times in seconds, and speedup.

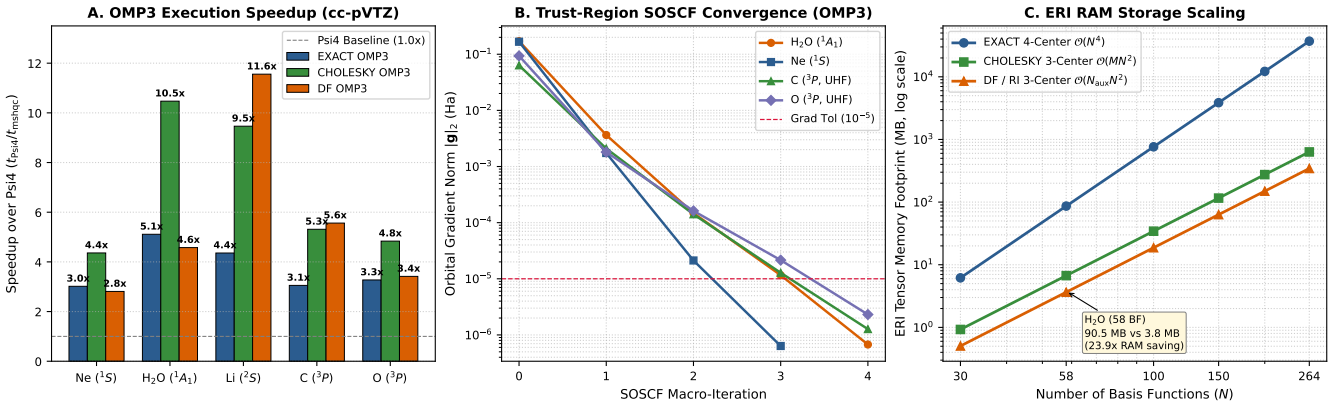


Figure 1: Benchmark evaluation of `mshqc` vs `Psi4` in the `cc-pVTZ` (`cc-pVTZ-RI`) basis: (A) `OMP3` wall-clock speedup ($t_{\mathrm{Psi4}}/t_{\mathrm{mshqc}}$) across `exact`, `cholesky`, and `df` integral modes; (B) Trust-Region SOSCF orbital gradient norm $\|\mathbf{g}\|_2$ convergence per macro-iteration for closed- and open-shell systems.

System	ERI	Method	Energy (Ha)	$\lvert\Delta E\rvert$	`mshqc` (s)	`Psi4` (s)	Speedup
Ne (^1S)	exact	RHF	-128.53186164	0.00	0.200	0.749	3.75x
Ne (^1S)	exact	OMP2	-128.81022310	0.00	0.093	0.226	2.43x
Ne (^1S)	exact	OMP3	-128.81027068	0.00	0.117	0.353	3.02x
Ne (^1S)	cholesky	OMP3	-128.81027068	0.00	0.125	0.545	4.36x
Ne (^1S)	df	OMP3	-128.81042564	0.00	0.116	0.326	2.81x
H_2O ($^1A_1$)	exact	RHF	-76.05712742	0.00	0.062	0.365	5.89x
H_2O ($^1A_1$)	exact	OMP2	-76.33406054	0.00	0.496	1.569	3.16x
H_2O ($^1A_1$)	exact	OMP3	-76.33711056	0.00	0.797	4.074	5.11x
H_2O ($^1A_1$)	cholesky	OMP2	-76.33406044	0.10	0.330	3.146	9.53x
H_2O ($^1A_1$)	cholesky	OMP3	-76.33711183	1.27	0.705	7.382	10.47x
H_2O ($^1A_1$)	df	RMP2	-76.33220605	0.00	0.005	0.066	13.20x
H_2O ($^1A_1$)	df	OMP2	-76.33402331	0.00	0.144	0.698	4.85x
H_2O ($^1A_1$)	df	OMP3	-76.33725847	0.00	0.358	1.638	4.58x
Li (^2S, UHF)	exact	OMP2	-7.44391221	0.00	0.033	0.572	17.33x
Li (^2S, UHF)	cholesky	OMP3	-7.44569499	0.00	0.149	1.410	9.46x
Li (^2S, UHF)	df	OMP3	-7.44569708	0.00	0.104	1.202	11.56x
C (^3P, UHF)	exact	OMP3	-37.78269153	0.00	0.299	0.913	3.05x
C (^3P, UHF)	cholesky	OMP3	-37.78269153	0.00	0.291	1.546	5.31x
C (^3P, UHF)	df	OMP2	-37.76763043	0.00	0.055	0.471	8.56x
C (^3P, UHF)	df	OMP3	-37.78281240	0.00	0.192	1.068	5.56x
O (^3P, UHF)	exact	OMP3	-74.98104071	0.00	0.297	0.972	3.27x
O (^3P, UHF)	cholesky	OMP2	-74.96657683	0.00	0.092	0.818	8.89x
O (^3P, UHF)	cholesky	OMP3	-74.98104071	0.00	0.360	1.741	4.84x
O (^3P, UHF)	df	OMP2	-74.96656224	0.00	0.044	0.533	12.11x
O (^3P, UHF)	df	OMP3	-74.98111725	0.00	0.312	1.066	3.42x

Availability and documentation

`mshqc` is released under the Apache License 2.0 and hosted on GitHub (https://github.com/syahrulhidayat/mshqc), with precompiled Linux wheels built via automated GitHub Actions workflows. Documentation, quickstart tutorials, and troubleshooting guides are included in the `docs/` directory of the repository.

AI usage disclosure

During the preparation of this work, the author used generative AI tools (Gemini) to assist with code refactoring, performance profiling analysis, and manuscript editing. The author reviewed, validated, and tested all code and text against analytical quantum chemistry references and assumes full responsibility for the published work.

Acknowledgements

The author acknowledges the developers of `libcint` (Sun 2015) for analytical Gaussian integral evaluation and `TBLIS` (Matthews 2017) for transpose-free tensor contractions, as well as the `PySCF` (Sun et al. 2020) and `Psi4` (Smith et al. 2020) projects for providing reference values during numerical validation.

References

Atkins, P. W., M. S. Child, and C. S. G. Phillips. 1970. *Tables for Group Theory.* Oxford: Oxford University Press.

Beebe, Nelson H. F., and Jan Linderberg. 1977. "Simplifications in the Generation and Transformation of Two-Electron Integrals in Molecular Calculations." *International Journal of Quantum Chemistry* 12 (4): 683–705. https://doi.org/10.1002/qua.560120408.

Bozkaya, Uğur. 2011. "Orbital-Optimized Third-Order Møller–Plesset Perturbation Theory and Its Spin-Component and Spin-Opposite Scaled Variants: Application to Symmetry-Breaking Problems." *The Journal of Chemical Physics* 135 (22): 224103. https://doi.org/10.1063/1.3665134.

Bozkaya, Uğur, Justin M. Turney, Yukio Yamaguchi, Henry F. Schaefer, and C. David Sherrill. 2011. "Quadratically Convergent Algorithm for Orbital Optimization in the Orbital-Optimized Coupled-Cluster Doubles Method and in Orbital-Optimized Second-Order Møller-Plesset Perturbation Theory." *The Journal of Chemical Physics* 135 (10): 104103. https://doi.org/10.1063/1.3631129.

Dagum, Leonardo, and Ramesh Menon. 1998. "OpenMP: An Industry Standard API for Shared-Memory Programming." *IEEE Computational Science and Engineering* 5 (1): 46–55. https://doi.org/10.1109/99.660313.

Dunning, Thom H., Jr. 1989. "Gaussian Basis Sets for Use in Correlated Molecular Calculations. I. The Atoms Boron Through Neon and Hydrogen." *The Journal of Chemical Physics* 90 (2): 1007–23. https://doi.org/10.1063/1.456153.

Feyereisen, Martin, George Fitzgerald, and Andrew Komornicki. 1993. "Use of Approximate Integrals in Ab Initio Theory. Anick-Type Second Order Møller–Plesset Perturbation Theory." *Chemical Physics Letters* 208 (5-6): 359–63. https://doi.org/10.1016/0009-2614(93)87156-W.

Guennebaud, Gaël, Benoît Jacob, et al. 2010. "Eigen V3." https://eigen.tuxfamily.org.

Häser, Marco, and Reinhart Ahlrichs. 1989. "Improvements on the Direct SCF Method." *Journal of Computational Chemistry* 10 (1): 104–11. https://doi.org/10.1002/jcc.540100111.

Head-Gordon, Martin, John A. Pople, and Michael J. Frisch. 1988. "MP2 Energy Evaluation by Direct Methods." *Chemical Physics Letters* 153 (6): 503–6. https://doi.org/10.1016/0009-2614(88)85250-3.

Helgaker, Trygve, Poul Jørgensen, and Jeppe Olsen. 2000. *Molecular Electronic-Structure Theory.* Chichester: John Wiley & Sons. https://doi.org/10.1002/9781119019572.

Jakob, Wenzel. 2022. "Nanobind: Tiny and Efficient C++/Python Bindings." https://github.com/wjakob/nanobind.

Koch, Henrik, Alfredo Sánchez de Merás, and Thomas Bondo Pedersen. 2003. "Reduced Scaling in Electronic Structure Calculations Using Cholesky Decompositions." *The Journal of Chemical Physics* 118 (21): 9481–84. https://doi.org/10.1063/1.1578621.

Matthews, Devin A. 2017. "High-Performance Tensor Contraction Without Transposition." *SIAM Journal on Scientific Computing* 39 (6): C476–509. https://doi.org/10.1137/16M108968X.

Møller, Chr., and Milton S. Plesset. 1934. "Note on an Approximation Treatment for Many-Electron Systems." *Physical Review* 46 (7): 618–22. https://doi.org/10.1103/PhysRev.46.618.

Neese, Frank, Tobias Schwabe, Simone Kossmann, Bernd Schirmer, and Stefan Grimme. 2009. "Assessment of Orbital-Optimized, Spin-Component Scaled Second-Order Many-Body Perturbation Theory for Thermochemistry and Kinetics." *Journal of Chemical Theory and Computation* 5 (11): 3060–73. https://doi.org/10.1021/ct9003299.

Pople, John A., J. Stephen Binkley, and Rolf Seeger. 1976. "Theoretical Models Incorporating Electron Correlation." *International Journal of Quantum Chemistry* 10 (S10): 1–19. https://doi.org/10.1002/qua.560100802.

Pople, John A., and Robert K. Nesbet. 1954. "Self-Consistent Orbitals for Radicals." *The Journal of Chemical Physics* 22 (3): 571–72. https://doi.org/10.1063/1.1740120.

Pulay, Péter. 1980. "Convergence Acceleration of Iterative Sequences. The Case of SCF Iteration." *Chemical Physics Letters* 73 (2): 393–98. https://doi.org/10.1016/0009-2614(80)80396-4.

———. 1982. "Improved SCF Convergence Acceleration." *Journal of Computational Chemistry* 3 (4): 556–60. https://doi.org/10.1002/jcc.540030413.

Roothaan, Clemens C. J. 1951. "New Developments in Molecular Orbital Theory." *Reviews of Modern Physics* 23 (2): 69–89. https://doi.org/10.1103/RevModPhys.23.69.

———. 1960. "Self-Consistent Field Theory for Open Shells of Electronic Systems." *Reviews of Modern Physics* 32 (2): 179–85. https://doi.org/10.1103/RevModPhys.32.179.

Smith, Daniel G. A., Lori A. Burns, Andrew C. Simmonett, Robert M. Parrish, Matthew C. Schieber, Raimondas Galvelis, Peter Kraus, et al. 2020. "Psi4 1.4: Open-Source Software for High-Throughput Quantum Chemistry." *The Journal of Chemical Physics* 152 (18): 184108. https://doi.org/10.1063/5.0006002.

Sun, Qiming. 2015. "Libcint: An Efficient General Integral Library for Gaussian Basis Functions." *Journal of Computational Chemistry* 36 (22): 1664–71. https://doi.org/10.1002/jcc.23981.

Sun, Qiming, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, et al. 2020. "Recent Developments in the PySCF Program Package." *The Journal of Chemical Physics* 153 (2): 024109. https://doi.org/10.1063/5.0006074.

The HDF Group. 1997. "Hierarchical Data Format, Version 5." https://www.hdfgroup.org/HDF5/.

Weigend, Florian. 2002. "A Fully Direct RI-HF Algorithm: Implementation, Optimised Auxiliary Basis Sets, Demonstration of Accuracy and Efficiency." *Physical Chemistry Chemical Physics* 4 (18): 4285–91. https://doi.org/10.1039/B204199P.